10 **(a)** A student observes different stars from the Earth.
Give **two** reasons why some stars appear brighter than others.

1 .....

.....

2 .....

.....
[2]

(b) State what is meant by a standard candle.

.....

..... [1]

(c) A spectral line from a star within a galaxy is observed to have a wavelength of 660.9 nm. The same spectral line measured in the laboratory is observed to have a wavelength of 656.3 nm.

(i) Show that the speed of the star relative to the Earth is $2.1 \times 10^{6}\,\text{m}\,\text{s}^{-1}$.

[1]

(ii) Calculate the distance to the star.

The Hubble constant is $2.3 \times 10^{-18}\,\text{s}^{-1}$.

distance = ... m [2]

(iii) State and explain what can be concluded about the Universe based on this change in observed wavelength.

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.....

.....

..... [3]

[Total: 9]

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